

CANDIDATE
NAME

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CENTRE
NUMBER

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CANDIDATE
NUMBER

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MATHEMATICS

Paper 2 (Extended)

0580/22

May/June 2018

1 hour 30 minutes

Candidates answer on the Question Paper.

Additional Materials:

Electronic calculator

Geometrical instruments

Tracing paper (optional)

READ THESE INSTRUCTIONS FIRST

Write your Centre number, candidate number and name on all the work you hand in.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

DO NOT WRITE IN ANY BARCODES.

Answer **all** questions.

If working is needed for any question it must be shown below that question.

Electronic calculators should be used.

If the degree of accuracy is not specified in the question, and if the answer is not exact, give the answer to three significant figures. Give answers in degrees to one decimal place.

For π , use either your calculator value or 3.142.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

The total of the marks for this paper is 70.

This document consists of **15** printed pages and **1** blank page.

- 1 One morning, Marcia works from 08 20 to 11 15.

Find how long she works for.

Give your answer in hours and minutes.

..... h min [1]

- 2 Expand.

$$7(x - 8)$$

..... [1]

- 3 Here is a sequence.

a , 13, 9, 3, -5, -15, b , ...

Find the value of a and the value of b .

$a =$

$b =$ [2]

- 4 Complete these statements.

(a) When $w =$, $10w = 70$. [1]

(b) When $5x = 15$, $12x =$ [1]

- 5 22 17 25 41 39 4

Work out the difference between the two prime numbers in the list above.

..... [2]

- 6 Without using your calculator, work out $\frac{2}{3} - \frac{1}{12}$.

You must show all your working and give your answer as a fraction in its simplest form.

..... [2]

- 7 A and B are two towns on a map.
The bearing of A from B is 140° .

Work out the bearing of B from A .

..... [2]

- 8 Here are some numbers written in standard form.

$$3.4 \times 10^{-1} \quad 1.36 \times 10^6 \quad 7.9 \times 10^0 \quad 2.4 \times 10^5 \quad 5.21 \times 10^{-3} \quad 4.3 \times 10^{-2}$$

From these numbers, write down

- (a) the largest number,

..... [1]

- (b) the smallest number.

..... [1]

- 9 Using a straight edge and compasses only, construct the locus of points that are equidistant from A and B .

$A \bullet$

$\bullet B$

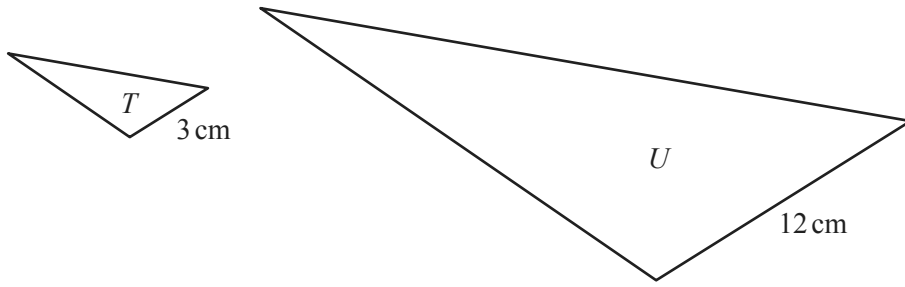
[2]

- 10 Factorise completely.

$$xy + 2y + 3x + 6$$

..... [2]

11

NOT TO
SCALE

The diagram shows two mathematically similar triangles, T and U .
Two corresponding side lengths are 3 cm and 12 cm.
The area of triangle T is 5 cm^2 .

Find the area of triangle U .

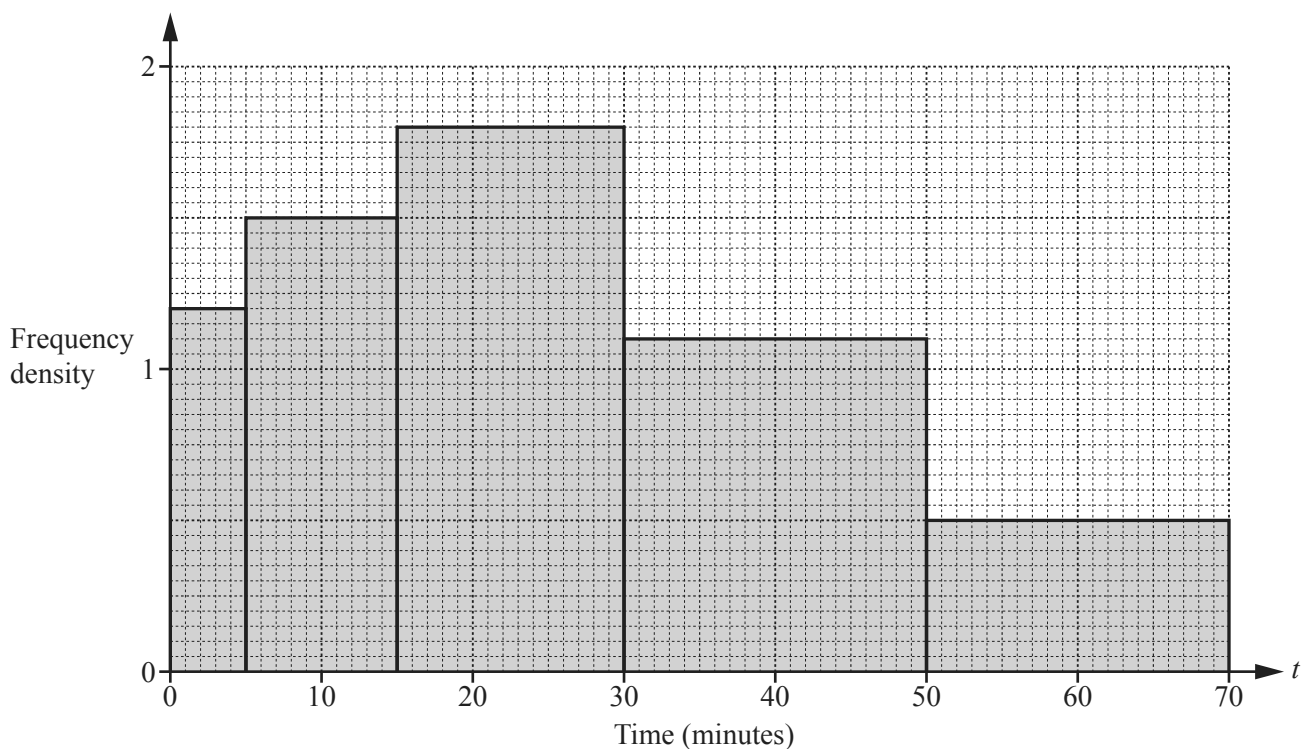
..... cm^2 [2]

- 12 Anna walks 31 km at a speed of 5 km/h.
Both values are correct to the nearest whole number.

Work out the upper bound of the time taken for Anna's walk.

..... hours [2]

- 13 The histogram shows information about the time, t minutes, spent in a shop by each of 80 people.

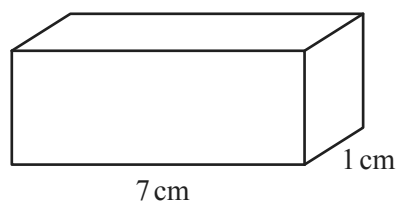


Complete the frequency table.

Time (t minutes)	$0 < t \leq 5$	$5 < t \leq 15$	$15 < t \leq 30$	$30 < t \leq 50$	$50 < t \leq 70$
Number of people	6		27		10

[2]

14



NOT TO
SCALE

The diagram shows a solid cuboid with base area 7 cm^2 .
The volume of this cuboid is 21 cm^3 .

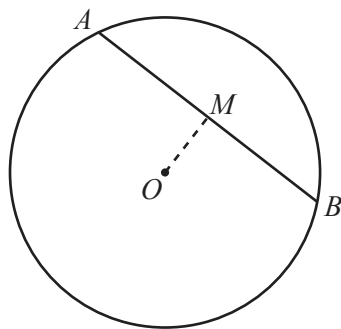
Work out the total surface area.

..... cm^2 [3]

- 15 Find the volume of a cylinder of radius 5 cm and height 8 cm.
Give the units of your answer.

..... [3]

16



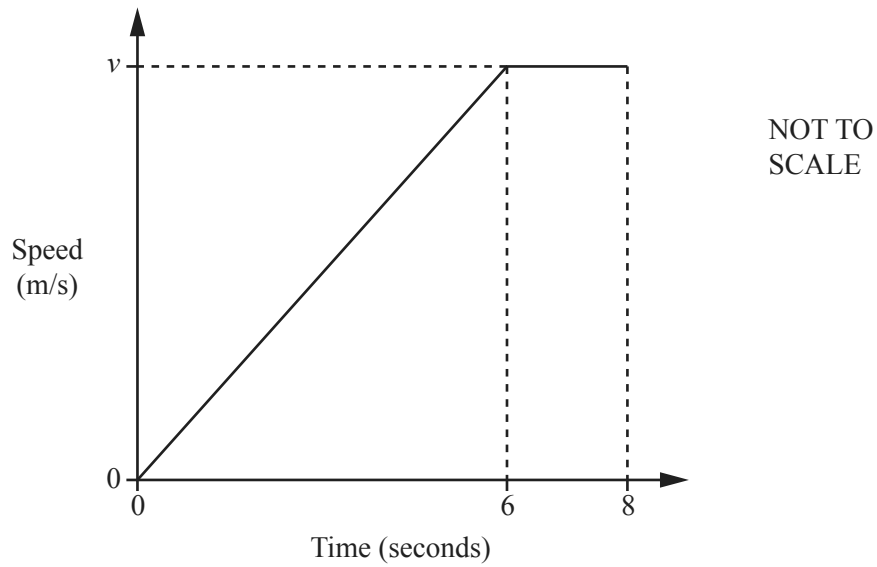
NOT TO
SCALE

The diagram shows a circle, centre O .
 AB is a chord of length 12 cm.
 M is the mid-point of AB and $OM = 4.5$ cm.

Calculate the radius of the circle.

..... cm [3]

- 17 The diagram shows information about the first 8 seconds of a car journey.



The car travels with constant acceleration reaching a speed of v m/s after 6 seconds.
The car then travels at a constant speed of v m/s for a further 2 seconds.
The car travels a total distance of 150 metres.

Work out the value of v .

$v = \dots\dots\dots$ [3]

- 18 A ball falls d metres in t seconds.
 d is directly proportional to the square of t .
 The ball falls 44.1 m in 3 seconds.

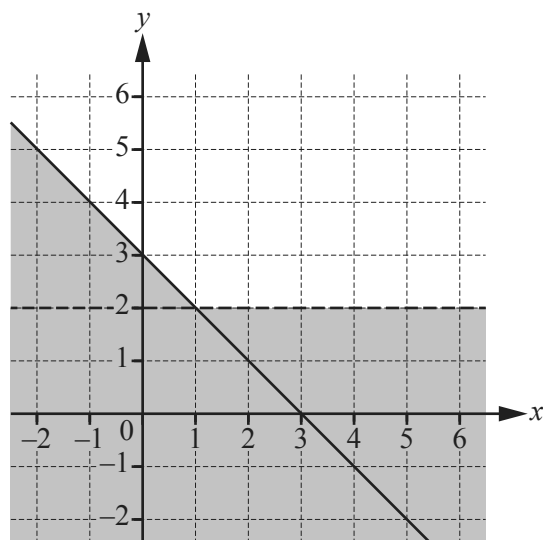
(a) Find a formula for d in terms of t .

$d = \dots\dots\dots$ [2]

(b) Calculate the distance the ball falls in 2 seconds.

$\dots\dots\dots$ m [1]

19



Find the two inequalities that define the region on the grid that is **not** shaded.

$\dots\dots\dots$
 $\dots\dots\dots$ [3]

20 $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 9 & 9 \end{pmatrix}$ $\mathbf{B} = \begin{pmatrix} 0 & 1 \\ 9 & 8 \end{pmatrix}$ $\mathbf{C} = \begin{pmatrix} 1 & 1 \\ 3 & 3 \end{pmatrix}$ $\mathbf{I} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

(a) Here are four matrix calculations.

$\mathbf{AI}$

$\mathbf{IA}$

$\mathbf{C}^2$

$\mathbf{B} + \mathbf{I}$

Work out which matrix calculation does **not** give the answer $\begin{pmatrix} 1 & 1 \\ 9 & 9 \end{pmatrix}$.

..... [2]

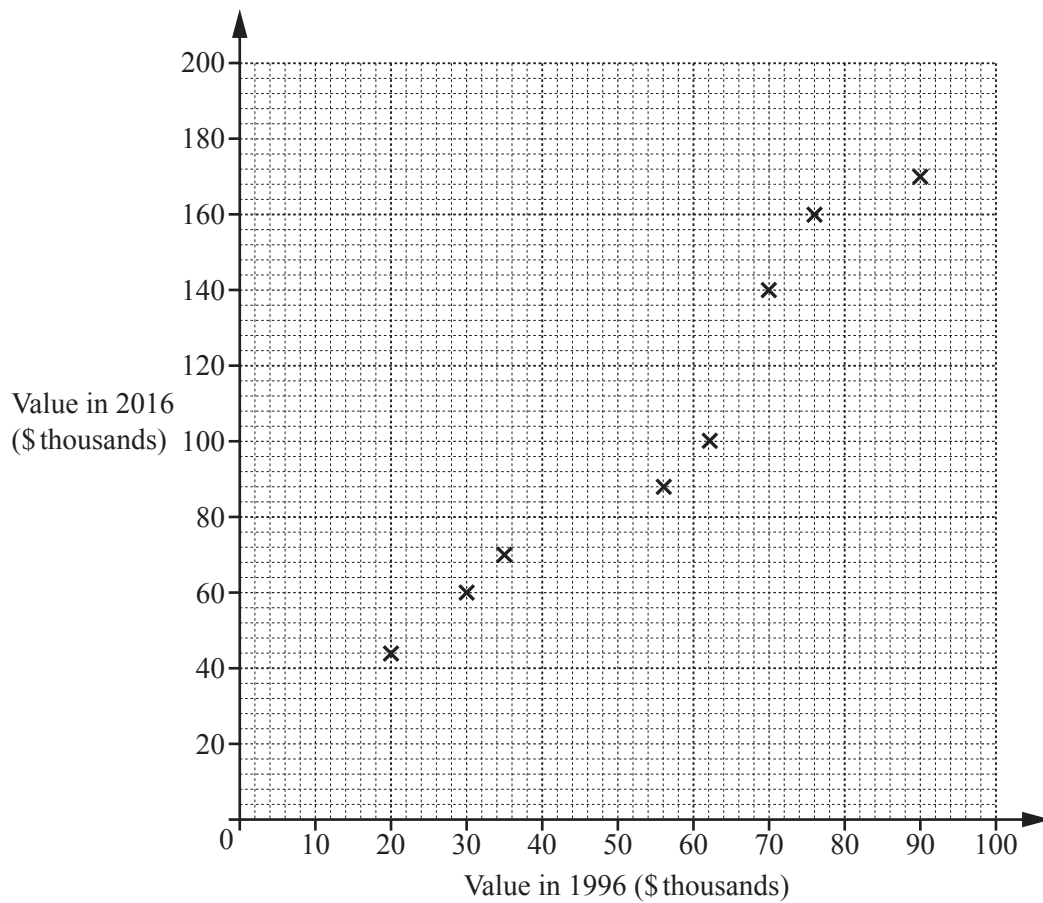
(b) Find $|\mathbf{B}|$.

..... [1]

(c) Explain why matrix $\mathbf{A}$ has no inverse.

..... [1]

- 21 The scatter diagram shows the value, in thousands of dollars, of eight houses in 1996 and the value of the same houses in 2016.



- (a) One of these eight houses had a value of \$70 000 in 1996.

Write down the value of this house in 2016.

\$ [1]

- (b) The values of two more houses are shown in the table.

Value in 1996 (\$ thousands)	40	80
Value in 2016 (\$ thousands)	80	150

On the scatter diagram, plot these values.

[1]

- (c) On the scatter diagram, draw a line of best fit.

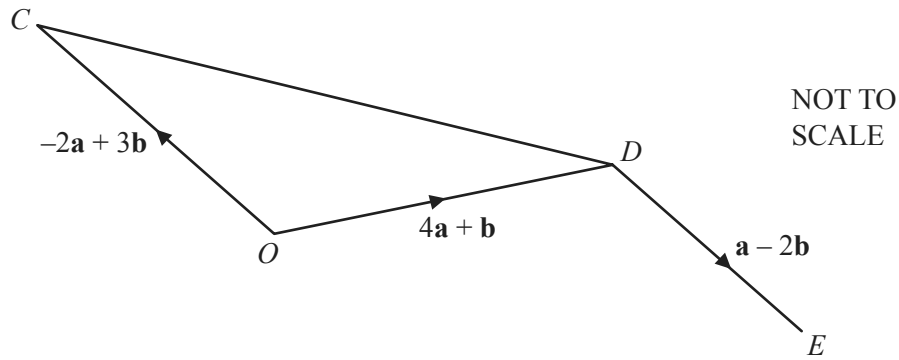
[1]

- (d) Another house had a value of \$50 000 in 1996.

Find an estimate of the value of this house in 2016.

\$ [1]

22



In the diagram, O is the origin, $\overrightarrow{OC} = -2\mathbf{a} + 3\mathbf{b}$ and $\overrightarrow{OD} = 4\mathbf{a} + \mathbf{b}$.

- (a) Find $\overrightarrow{CD}$, in terms of $\mathbf{a}$ and $\mathbf{b}$, in its simplest form.

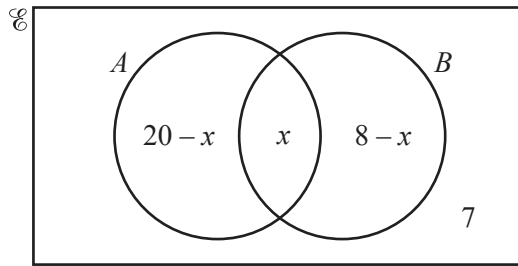
$$\overrightarrow{CD} = \dots\dots\dots [2]$$

- (b) $\overrightarrow{DE} = \mathbf{a} - 2\mathbf{b}$

Find the position vector of E , in terms of $\mathbf{a}$ and $\mathbf{b}$, in its simplest form.

$$\dots\dots\dots [2]$$

- 23 The Venn diagram shows information about the number of elements in sets A , B and $\mathcal{C}$.



- (a) $n(A \cup B) = 23$

Find the value of x .

$x = \dots\dots\dots$ [3]

- (b) An element is chosen at random from $\mathcal{C}$.

Find the probability that this element is in $(A \cup B)'$.

$\dots\dots\dots$ [2]

- 24** Box A and box B each contain blue and green pens only.
Raphael picks a pen at random from box A and Paulo picks a pen at random from box B .
The probability that Raphael picks a blue pen is $\frac{2}{3}$.
The probability that both Raphael and Paulo pick a blue pen is $\frac{8}{15}$.

(a) Find the probability that Paulo picks a blue pen.

..... [2]

(b) Find the probability that both Raphael and Paulo pick a green pen.

..... [3]

25 P is the point $(16, 9)$ and Q is the point $(22, 24)$.

- (a)** Find the equation of the line perpendicular to PQ that passes through the point $(5, 1)$.
Give your answer in the form $y = mx + c$.

$y = \dots\dots\dots$ [4]

- (b)** N is the point on PQ such that $PN = 2NQ$.

Find the co-ordinates of N .

$(\dots\dots\dots, \dots\dots\dots)$ [2]

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